# Scoring Matrices

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# Introduction

- ❑Scoring matrices are tools used in bioinformatics to quantify the similarity or difference between sequences of amino acids (proteins) or nucleotides (DNA/RNA).
- ❑They are crucial in sequence alignment algorithms like BLAST, FASTA, and others to evaluate whether two sequences are evolutionarily or functionally related.
- ❑<u>Key Types of Amino acid Scoring Matrices (Empirical studies):</u>

→PAM (Point Accepted Mutation)

→BLOSUM (Blocks Substitution Matrix)

# Basics

- Transition: purine replaced by purine or pyrimidine replaced by pyrimidine
- Transversion: purine by pyrimidine or vice versa.
- Transition mutation’s frequency is more than transversion mutation
- Model required to reflect the probability of mutations
- Scoring of AA based on physiochemical properties. So, AA with similar physiochemical properties can substitute each other easily.

→ Aromatic AA: phenylalanine, tryptophan and tyrosine.
→ Basic AA: Lysine, Arginine, Histidine.
→ Acid AA: Aspartic acid, glutamic acid, asparagine, glutamine
→ Hydrophobic AA: Methionine, isoleucine, leucine, valine
→ Polar AA: serine, threonine and cystine

- Cystine has sulfhydryl group for metal binding, active sites and disulfide bond formation. When cystine is substituted by other AA then the enzymatic property or the structure may disrupt.
- Glycine and proline: responsible for structure of protein. Substitution of these AAs may lead to change on function

→ e.g., In insulin, cysteine residues form disulfide bonds that are critical for the hormone's functional conformation. If cysteine is replaced, insulin may lose its structure and no longer be able to bind to its receptor, impairing glucose regulation.
→ In enzymes like caspases (involved in apoptosis), substitution of the catalytic cysteine would disable their function, preventing apoptosis and potentially leading to unregulated cell growth (e.g., cancer).

# One Letter Code for AA

- Alanine: A
- Arginine: R
- Asparagine: N
- Aspartic acid: D
- Cysteine: C
- Glutamine: Q
- Glutamic acid: E
- Glycine: G
- Histidine: H
- Isoleucine: I
- Leucine: L
- Lysine: K
- Methionine: M
- Phenylalanine: F
- Proline: P
- Serine: S
- Threonine: T
- Tryptophan: W
- Tyrosine: Y
- Valine: V

# PAM

- A **point accepted mutation** — also known as a PAM — is the replacement of a single amino acid in the primary structure of a protein with another single amino acid, which is accepted by the processes of natural selection.
- This definition does not include all point mutations in the DNA of an organism.
- These matrices are based on observed mutations in sequences that have evolved from a common ancestor, representing the probability of one amino acid being replaced by another over evolutionary time.
- A PAM matrix is a matrix where each column and row represents one of the twenty standard amino acids.
- In bioinformatics, PAM matrices are regularly used as substitution matrices to score sequence alignments for proteins.
- Each entry in a PAM matrix indicates the likelihood of the amino acid of that row being replaced with the amino acid of that column.

# PAM

- PAM matrices were introduced by Margaret Dayhoff in 1978 (History):

→ **Collection of Protein Sequence Data:**

The goal was to analyze these sequences to understand the patterns of amino acid substitutions that occur over evolutionary time.

→ **Identification of Closely Related Sequences & Constructed phylogenetic tree:**

The calculation of these matrices were based on 1572 observed mutations in the phylogenetic trees of 71 families of closely related proteins.

The proteins to be studied were selected on the basis of having high similarity.

The protein alignments included were required to display at least 85% identity.

→ **Observation of Mutations:**

By comparing these homologous protein sequences, they observed how amino acids were substituted over time. They tracked accepted mutations, which are substitutions that have occurred in the evolutionary history of the protein.

# PAM

- PAM matrices were introduced by Margaret Dayhoff in 1978 (History):

→ **Calculation of Substitution Frequencies/Amino acid replacement (Log Odds Ratio)**

Calculation of the frequencies of different amino acid substitutions. For each amino acid pair, they determined how often one amino acid was replaced by another in the evolution of the protein sequences.

→ **Generation of the PAM1 Matrix**

Represents the substitutions that are expected after 1% divergence from a common ancestor, or approximately 1 accepted mutation per 100 amino acids. Used when comparing sequences that are highly similar or have diverged recently.

→ **Extrapolation to Other PAM Matrices (PAM250)**

Represents a much greater evolutionary time, with 250% divergence, where a sequence has undergone 250 substitutions for every 25,000 amino acids. It implies a more distant relationship between sequences. Used for sequences that are more distantly related and have evolved over a longer time.

→ **Validation and Refinement**

# PAM

- **Observed Substitution Frequencies:**

First, the observed frequencies of amino acid substitutions are collected from known protein sequences that have undergone mutations. These are typically homologous sequences, meaning they have a common evolutionary origin. For each pair of amino acids (e.g., Alanine (A) to Glycine (G)), researchers count how often one amino acid is substituted by another.

- **Expected Substitution Frequencies:**

To calculate the expected substitution frequencies, researchers consider the probability of amino acids being substituted by random chance. This is based on the overall frequency of each amino acid in proteins. For instance, if amino acids occur randomly, the probability of substitution between amino acids is based on how common they are in the protein world.

# PAM

- **<u>Logarithmic Conversion (Log Odds Ratio):</u>**

The PAM matrix scores are calculated using a logarithmic transformation of the ratio of observed to expected substitution frequencies. This is known as the log odds ratio and is calculated as:

$$\text{Score} = \log\left(\frac{\text{Observed frequency of substitution}}{\text{Expected frequency of substitution (random chance)}}\right)$$

- This log ratio is important because it quantifies how much more or less likely an observed substitution is compared to what we would expect by random chance. Here's how the scores work:

→ **Positive score (+ve)**: The substitution occurs **more frequently** than expected by random chance, often between amino acids with similar physicochemical properties (e.g., substitution between **Leucine (L)** and **Isoleucine (I)**).

→ **Zero score (0)**: The substitution occurs at a frequency **equal to random chance**, meaning no specific evolutionary preference for that substitution.

→ **Negative score (-ve)**: The substitution occurs **less frequently** than random chance, often between amino acids with **dissimilar physicochemical properties** (e.g., substitution between **Aspartic acid (D)** and **Leucine (L)**, which have different sizes and charges).

# PAM

- Each PAM matrix has twenty rows and twenty columns — one representing each of the twenty amino acids translated by the genetic code.
- The value in each cell of a PAM matrix is related to the probability of a row amino acid before the mutation being aligned with a column amino acid afterwards.
- Replacement odd: The AA residues most likely to be replaced have the largest diagonal value
- Exchange odd: exchange values of AA getting exchange/substituted by another AA

Replacement Odd

Exchange Odd

| | C | S | T | P | A | G | N | D | E | Q | H | R | K | M | I | L | V | F | Y | W | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C | 12 | | | | | | | | | | | | | | | | | | | | C |
| S | 0 | 2 | | | | | | | | | | | | | | | | | | | S |
| T | −2 | 1 | 3 | | | | | | | | | | | | | | | | | | T |
| P | −3 | 1 | 0 | 6 | | | | | | | | | | | | | | | | | P |
| A | −2 | 1 | 1 | 1 | 2 | | | | | | | | | | | | | | | | A |
| G | −3 | 1 | 0 | −1 | 1 | 5 | | | | | | | | | | | | | | | G |
| N | −4 | 1 | 0 | −1 | 0 | 0 | 2 | | | | | | | | | | | | | | N |
| D | −5 | 0 | **0** | −1 | 0 | 1 | 2 | 4 | | | | | | | | | | | | | D |
| E | −5 | **0** | **0** | −1 | 0 | 0 | 1 | 3 | 4 | | | | | | | | | | | | E |
| Q | −5 | −1 | −1 | **0** | 0 | −1 | 1 | 2 | 2 | 4 | | | | | | | | | | | Q |
| H | −3 | −1 | −1 | 0 | −1 | −2 | 2 | 1 | 1 | 3 | 6 | | | | | | | | | | H |
| R | −4 | 0 | −1 | 0 | −2 | −3 | 0 | −1 | −1 | 1 | 2 | 6 | | | | | | | | | R |
| K | −5 | 0 | 0 | −1 | −1 | −2 | 1 | 0 | 0 | 1 | 0 | 3 | 5 | | | | | | | | K |
| M | −5 | −2 | −1 | −2 | −1 | −3 | −2 | −3 | −2 | −1 | −2 | 0 | 0 | 6 | | | | | | | M |
| I | −2 | −1 | 0 | −2 | −1 | −3 | −2 | −2 | −2 | −2 | −2 | −2 | −2 | 2 | 5 | | | | | | I |
| L | −6 | −3 | −2 | −3 | −2 | −4 | −3 | −4 | −3 | −2 | −2 | −3 | −3 | 4 | 2 | 6 | | | | | L |
| V | −2 | −1 | 0 | −1 | 0 | −1 | −2 | −2 | −2 | −2 | −2 | −2 | −2 | 2 | 4 | 2 | 4 | | | | V |
| F | −4 | −3 | −3 | −5 | −4 | −5 | −4 | −6 | −5 | −5 | −2 | −4 | −5 | 0 | 1 | 2 | −1 | 9 | | | F |
| Y | 0 | −3 | −3 | −5 | −3 | −5 | −2 | −4 | −4 | −4 | 0 | −4 | −4 | −2 | −1 | −1 | −2 | 7 | 10 | | Y |
| W | −8 | −2 | −5 | −6 | −6 | −7 | −4 | −7 | −7 | −5 | −3 | 2 | −3 | −4 | −5 | −2 | −6 | 0 | 0 | 17 | W |
| | C | S | T | P | A | G | N | D | E | Q | H | R | K | M | I | L | V | F | Y | W | |

# PAM

- (C) sulfhydryl
- (STPAG) small hydrophilic
- (NDEQ) acid, acid amide and hydrophilic
- (HRK) basic
- (MILV) small hydrophobic
- (FYW) aromatic.

| | C | S | T | P | A | G | N | D | E | Q | H | R | K | M | I | L | V | F | Y | W | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C | 12 | | | | | | | | | | | | | | | | | | | | C |
| S | 0 | 2 | | | | | | | | | | | | | | | | | | | S |
| T | −2 | 1 | 3 | | | | | | | | | | | | | | | | | | T |
| P | −3 | 1 | 0 | 6 | | | | | | | | | | | | | | | | | P |
| A | −2 | 1 | 1 | 1 | 2 | | | | | | | | | | | | | | | | A |
| G | −3 | 1 | 0 | −1 | 1 | 5 | | | | | | | | | | | | | | | G |
| N | −4 | 1 | 0 | −1 | 0 | 0 | 2 | | | | | | | | | | | | | | N |
| D | −5 | 0 | **0** | −1 | 0 | 1 | 2 | 4 | | | | | | | | | | | | | D |
| E | −5 | **0** | **0** | −1 | 0 | 0 | 1 | 3 | 4 | | | | | | | | | | | | E |
| Q | −5 | −1 | −1 | **0** | 0 | −1 | 1 | 2 | 2 | 4 | | | | | | | | | | | Q |
| H | −3 | −1 | −1 | 0 | −1 | −2 | 2 | 1 | 1 | 3 | 6 | | | | | | | | | | H |
| R | −4 | 0 | −1 | 0 | −2 | −3 | 0 | −1 | −1 | 1 | 2 | 6 | | | | | | | | | R |
| K | −5 | 0 | 0 | −1 | −1 | −2 | 1 | 0 | 0 | 1 | 0 | 3 | 5 | | | | | | | | K |
| M | −5 | −2 | −1 | −2 | −1 | −3 | −2 | −3 | −2 | −1 | −2 | 0 | 0 | 6 | | | | | | | M |
| I | −2 | −1 | 0 | −2 | −1 | −3 | −2 | −2 | −2 | −2 | −2 | −2 | −2 | 2 | 5 | | | | | | I |
| L | −6 | −3 | −2 | −3 | −2 | −4 | −3 | −4 | −3 | −2 | −2 | −3 | −3 | 4 | 2 | 6 | | | | | L |
| V | −2 | −1 | 0 | −1 | 6 | −1 | −2 | −2 | −2 | −2 | −2 | −2 | −2 | 2 | 4 | 2 | 4 | | | | V |
| F | −4 | −3 | −3 | −5 | −4 | −5 | −4 | −6 | −5 | −5 | −2 | −4 | −5 | 0 | 1 | 2 | −1 | 9 | | | F |
| Y | 0 | −3 | −3 | −5 | −3 | −5 | −2 | −4 | −4 | −4 | 0 | −4 | −4 | −2 | −1 | −1 | −2 | 7 | 10 | | Y |
| W | −8 | −2 | −5 | −6 | −6 | −7 | −4 | −7 | −7 | −5 | −3 | 2 | −3 | −4 | −5 | −2 | −6 | 0 | 0 | 17 | W |
| | C | S | T | P | A | G | N | D | E | Q | H | R | K | M | I | L | V | F | Y | W | |

# BLOSUM

- In bioinformatics, the **BLOSUM** (**BLO**cks **SU**bstitution **M**atrix) matrix is a substitution matrix used for sequence alignment of proteins.
- BLOSUM matrices were first introduced in a paper by Steven Henikoff and Jorja Henikoff.
- BLOSUM matrices are used to score alignments between evolutionarily divergent protein sequences. (Multiple alignment of distantly related proteins)
- They are based on local alignments.
- They scanned the BLOCKS database for very conserved regions of protein families (that do not have gaps in the sequence alignment) and then counted the relative frequencies of amino acids and their substitution probabilities.
- The scores in BLOSUM matrices reflect the probability of amino acid substitutions occurring in these highly conserved, ungapped alignment blocks.
- Then, they calculated a log-odds score for each of the 210 possible substitution pairs of the 20 standard amino acids.
- All BLOSUM matrices are based on observed alignments; they are not extrapolated from comparisons of closely related proteins like the PAM Matrices.

# BLOSUM

- <u>Steps:</u>

→ BLOCK sequences of related proteins were multiply aligned

→ Clustering approach was used while creating a BLOSUM matrix where sequences with 62% identity/similarity were clustered (BLOSUM62). Based on sequence similarity, BLOSUM matrix is named e.g., BLOSUM50, BLOSUM60

→ Calculate frequency of substitution of amino acids in the Block sequences

**Similarity** ⟷ **Difference**

PAM1
BLOSUM80

PAM250
BLOSUM50

# Difference between PAM & BLOSUM

| PAM | BLOSUM |
|---|---|
| To compare closely related sequences, matrices with lower numbers are created. | To compare closely related sequences, matrices with higher numbers are created. |
| To compare distantly related sequences, matrices with high numbers are created. | To compare distantly related sequences, matrices with low numbers are created. |
| Based on global alignments of closely related sequences. | Based on local alignments of closely related sequences. |
| PAM 1 is the matrix calculated from comparisons of sequences with no more than 1% divergence but corresponds to 99% sequence identity. | BLOSUM 62 is a matrix calculated from comparisons of sequences with a pairwise identity of no more than 62%. |
| Other PAM matrices are extrapolated from PAM 1. | Based on observed alignments; they are not extrapolated from comparisons of closely related sequences. |
| Higher numbers in matrices naming scheme denote larger evolutionary distance. | Larger numbers in matrices naming scheme denote higher sequence similarity and therefore smaller evolutionary distance. |

# Importance of PAM & BLOSUM

- **<u>Quantifying Similarity</u>**

- Even if sequences are similar, scoring matrices help to assign a **numerical score** to the similarity. Not all amino acid substitutions are equally likely or biologically meaningful. PAM and BLOSUM help by giving more weight to substitutions that are **biochemically favorable** and less weight (or negative scores) to rare or unlikely substitutions. This allows us to:
- **Distinguish functional or evolutionary relationships** from random similarities.
- Assess how strong the similarity is, which helps in comparing the likelihood of biological relevance.

- **<u>Accounting for Evolutionary Distance</u>**

- Two sequences might be similar, but **how similar?** Are they from **closely related species** or have they diverged over a **long evolutionary time**? Scoring matrices like **PAM** help account for the **evolutionary distance** between sequences:
- **PAM** matrices allow us to adjust for the expected number of mutations based on evolutionary time (e.g., PAM1 for close relationships, PAM250 for more distant).
- **BLOSUM** matrices are used to detect **functional similarity** and are designed to compare sequences that have evolved at different rates.
- Using these matrices helps **evaluate similarity more rigorously**, even when we already suspect a relationship.
- For example, replacing **phenylalanine (F)** with **tyrosine (Y)** might be acceptable because they are both **aromatic** amino acids, whereas substituting **phenylalanine (F)** with **glycine (G)** would be less favorable due to their significant differences in size and chemical properties. The matrices provide a meaningful score to reflect these differences.

# Importance of PAM & BLOSUM

- **<u>Optimizing Alignments:</u>**
- Scoring matrices like PAM and BLOSUM help identify the **best possible alignment** by rewarding substitutions that make biological sense and penalizing unlikely ones. This is especially important in tools like **BLAST** or **FASTA**, which aim to find the best alignment between sequences.

- **<u>Identifying Functional Similarity</u>**
- Two sequences might share overall similarity, but specific regions may have undergone **functional divergence**. PAM and BLOSUM allow us to zoom in on:
- **Conserved regions** that might be functionally important (e.g., active sites, binding regions).
- **Variable regions** that may have evolved due to environmental pressures or different functions.